# Combinatorial Optimization

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# Overview

- Linear/Integer Programming
- Totally Unimodular Matrices
- Matroids
- Gomory-Hu Trees
- Implementation of Gomory-Hu Tree in C++

**Linear Programs (LPs)**

$$\begin{aligned} \max\ & c^T x \\ \text{s.t. } & Ax \leq b, \\ & x \geq 0. \end{aligned}$$

Decision variable: $x$

Constraints: $Ax \leq b$, $x \geq 0$

Objective function: $c^T x$

# LP Duality

**Primal**

$$\max c^T x \quad \text{s.t. } Ax \le b,\quad x \ge 0.$$

**Dual**

$$\min b^T y \quad \text{s.t. } A^T y \ge c,\quad y \ge 0.$$

**Integer Programs (IPs)**

IPs are LPs such that any optimal solution must be integral.

$$\text{min dual IP} \leq \text{min dual LP} = \text{max primal LP} \leq \text{max primal IP}$$

Equality holds by duality.

**Totally Unimodular Matrix (TUM)**

A **totally unimodular matrix** is a matrix such that every square submatrix has determinant $-1$, 0, or 1.

In LP, if $A$ is TUM and $b$ and $c$ are integral, then

$$\min \text{dual IP} \leq \min \text{dual LP} = \max \text{primal LP} \leq \max \text{primal IP}$$

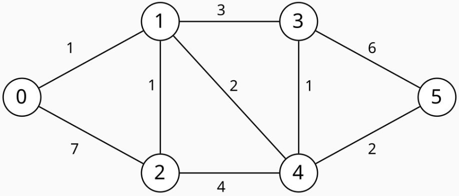
becomes

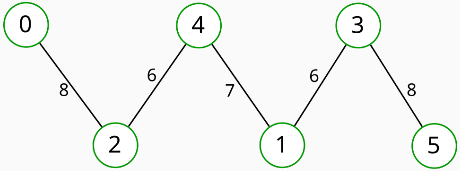
$$\min \text{dual IP} = \min \text{dual LP} = \max \text{primal LP} = \max \text{primal IP}$$

**Gomory-Hu Tree**

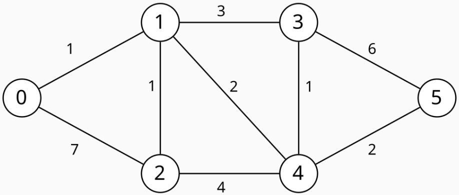
A **Gomory-Hu tree** of an undirected weighted graph is a weighted tree such that for every two vertices $s, t$ of a graph, the value of the min-cut equals the minimum edge weight on the unique $(s, t)$-path of the Gomory-Hu tree.

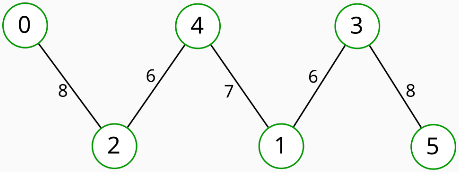
# Gomory-Hu Tree Example (0, 2)



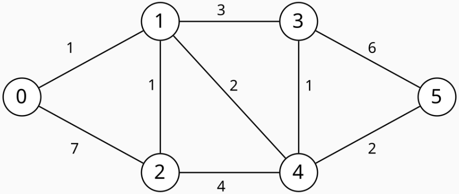
# Gomory-Hu Tree Example (0, 2)

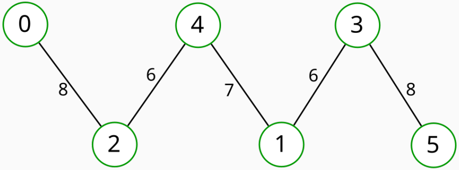


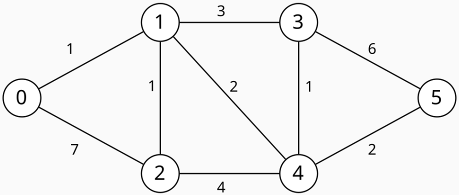

- One minimum $(0, 2)$-cut of original graph: $(0, 1), (0, 2) \implies 1 + 7 = 8.$
- Minimum-weight edge on $(0, 2)$-path of tree: $(0, 2) \implies 8.$

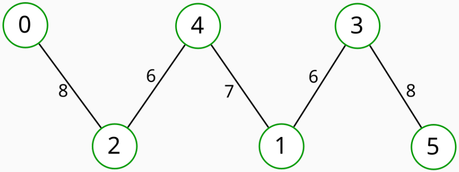
# Gomory-Hu Tree Example (2, 3)



# Gomory-Hu Tree Example (2, 3)




- One minimum $(2,3)$-cut of original graph: $(1,3),(3,4),(4,5) \implies 3+1+2=6$.
- Minimum-weight edge on $(2,3)$-path of tree: $(2,4)$ or $(1,3) \implies 6$.

Here is our GitHub repository with our program.

Thank you for listening to our presentation!